Pairwise edge correlations in random minimum spanning trees: a universal bound and complete-graph negative correlation

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Abstract

Let G be a finite connected multigraph whose edges receive independent weights from one atomless law, and let $\mathrm{MST}(G)$ be the resulting random minimum spanning tree. Its law is not pairwise negatively correlated: Lyons, Peres and Schramm exhibited two positively correlated edges, and we give such an example on a simple graph. We prove that the failure is nevertheless bounded uniformly:

$$\mathbf{P}_{\mathrm{MST}}(e, f \in T) \ \leq\ 8\, \mathbf{P}_{\mathrm{MST}}(e \in T)\, \mathbf{P}_{\mathrm{MST}}(f \in T),$$

answering a question of R. Lyons recorded by Tang and Zhang. After conditioning on all other weights, Harris's inequality gives conditional negative correlation; two bottleneck distances and a sharp second-moment estimate control the remaining environmental covariance. For K_n we prove pairwise negative correlation for every $n \geq 3$. The key finite identity is $\mathbf{E}[\deg(x)^2] = 10(n-1)/n - 4\mathbf{E}[L_n]$, where L_n is the total weight of the minimum spanning tree under rate-one exponential weights. Known expansions for $\mathbf{E}[L_n]$ then give the rate of convergence to $10 - 4\zeta(3)$ and the limits of both pair-correlation ratios. Finally, an explicit K_4 family shows that no universal constant survives when the independent edge laws need not be identical.

1 Introduction

Throughout, $G = (V, E)$ is a finite connected multigraph: loops and parallel edges are permitted. Each edge $g \in E$ receives a weight $w(g)$, and the weights are independent with one common atomless law μ on $\mathbb{R}$. Almost surely all weights are distinct, so the minimum spanning tree $T = \mathrm{MST}(G)$ is almost surely unique; we write $\mathbf{P}_{\mathrm{MST}}$ for its law, a probability measure on the spanning trees of G, equivalently on $\{0,1\}^E$. Since the minimum spanning tree depends on the weights only through their relative order, $\mathbf{P}_{\mathrm{MST}}$ does not depend on μ: uniform, exponential, or any other atomless law produce the same measure. We write

$$p_e = \mathbf{P}_{\mathrm{MST}}(e \in T), \qquad p_{ef} = \mathbf{P}_{\mathrm{MST}}(e \in T,\ f \in T).$$

Uniform spanning trees have negatively correlated edge indicators, so it is natural to ask whether the minimum spanning tree law behaves the same way; for the uniform spanning tree measure the pairwise inequality $p_{ef} \leq p_e p_f$ is classical [14, Chapter 4]. Following [19], say that $\mathbf{P}_{\mathrm{MST}}$ has the *pairwise negative correlation* (p-NC) property if $p_{ef} \leq p_e p_f$ for all $e \neq f$. The

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expectation is wrong. Lyons, Peres and Schramm [15, §5] take K_4, replace each of two disjoint edges by three parallel copies, and mark one copy in each bundle; then

$$p_e = p_f = \frac{331}{1260}, \qquad p_{ef} = \frac{109}{1575}, \qquad \frac{p_{ef}}{p_e p_f} = \frac{109872}{109561} = 1.00283\ldots > 1. \tag{1}$$

Thus the relevant quantity is the ratio $p_{ef}/(p_e p_f)$. We bound it uniformly over all graphs and determine its sign on the complete graph, where the naive inequality turns out to be true at every size.

1.1 A universal factor

Tang and Zhang [19, Question 3.13] record the following question, which they attribute to R. Lyons: is there a constant $C > 0$ such that $p_{ef} \le C\, p_e p_f$ for every finite connected graph G and every pair of distinct edges e, f of G? Their motivation is the analogous inequality with $C = 2$ for the *uniform forest* measure, which follows from the work of Brändén and Huh on Lorentzian polynomials [5]; for further uniform spanning subgraph measures see [20].

Theorem A. Let G be a finite connected multigraph whose edge weights are independent and identically distributed with an atomless law, and let $e \neq f$ be edges of G. Then

$$\mathbf{P}_{\mathrm{MST}}(e, f \in T) \;\le\; 8\, \mathbf{P}_{\mathrm{MST}}(e \in T)\, \mathbf{P}_{\mathrm{MST}}(f \in T).$$

Theorem A is proved in §3. It is stated for multigraphs, which is formally stronger than the question as posed and in particular covers the example (1); loops, bridges, and pairs of parallel marked edges are included rather than excluded, as §3.5 records.

The constant 8 is not optimal. Proposition 4.2 finds the exact maximum $78100/77841 = 1.00332\ldots$ in the two-parameter parallel-bundle family containing (1), but simple graphs already do better: Proposition 4.4 gives the ratio $13938405/13872419 = 1.00475\ldots$. Thus the optimal constant lies in $[13938405/13872419, 8]$. Corollary 3.5 improves the upper factor to $2 + o(1)$ when both marginals vanish.

Positive correlation is not an artefact of parallel edges. Proposition 4.3 gives a simple graph on five vertices with $p_{ef}/(p_e p_f) = 1450/1449 > 1$.

1.2 The complete graph, at every size

For K_n with $n \ge 3$ every edge has the same inclusion probability

$$p_0 = \mathbf{P}\big(e \in \mathrm{MST}(K_n)\big) = \frac{n-1}{\binom{n}{2}} = \frac{2}{n}, \tag{2}$$

because every spanning tree has $n-1$ edges and the edges of K_n are equivalent under automorphisms. Write p_1 for the probability that two fixed edges sharing a vertex both lie in the tree, and p_2 for the same probability for two fixed disjoint edges (defined for $n \ge 4$).

Tang and Zhang [19, Theorem 1.1] prove that $p_2 \le p_0^2$ for all n beyond an unspecified threshold, by running Fatou's lemma against the local weak limit of $\mathrm{MST}(K_n)$ [1]; their argument yields no value for the threshold. For adjacent pairs they reduce the problem to an upper bound on $\mathbf{E}[\deg(x)^2]$ and leave it open, writing that while the matching lower bound "via Fatou's lemma and the local limit ... is straightforward, proving a corresponding upper bound seems more challenging" [19, §3.2.2]. Both cases in fact hold at every size, strictly.

Theorem B1. For every $n \geq 3$ and every two distinct edges of K_n sharing a vertex,

$$p_1 \ < \ p_0^2 = \frac{4}{n^2}.$$

Theorem B2. For every $n \geq 4$ and every two disjoint edges of K_n,

$$p_2 \ < \ p_0^2 = \frac{4}{n^2}.$$

Corollary B3. For every $n \geq 3$ the measure $\mathbf{P}_{\mathrm{MST}}$ on K_n has the full pairwise negative correlation property. (For $n = 3$ there is no disjoint pair, so the statement there is exactly Theorem B1.)

These are proved in §5. Theorem B1 settles the adjacent case left open in [19, §3.2.2]. Theorem B2 gives an effective complete-range proof of the disjoint conclusion of [19, Theorem 1.1], by a different method and without local weak convergence or Fatou's lemma. The two inequalities are strict. In the computed range, p_1/p_0^2 increases from 0.7500 at $n = 3$ to 0.7851 at $n = 30$, and p_2/p_0^2 from 0.9778 at $n = 4$ to 0.9948 at $n = 30$ (Table 1). Strictness is a feature of K_n and not of the measure in general: if e and f are two bridges of a connected graph then $p_e = p_f = p_{ef} = 1$, and $p_{ef} = p_e p_f$ exactly.

1.3 An exact identity for the complete graph

Theorems B1 and B2 are two sides of one estimate, and the object they estimate is classical. Let L_n denote the total weight of $\mathrm{MST}(K_n)$ when the edge weights are i.i.d. Exp(1).

Proposition D1. For every $n \geq 2$ and every vertex x of K_n,

$$\mathbf{E}_{\mathrm{MST}(K_n)}[\deg(x)^2] = \frac{10(n-1)}{n} - 4\,\mathbf{E}[L_n].$$

Corollary D2. For $n \geq 3$, adjacent pairwise negative correlation on K_n is equivalent to the first estimate below; for $n \geq 4$, disjoint pairwise negative correlation is equivalent to the second:

$$p_1 \leq p_0^2 \iff \mathbf{E}[L_n] \ \geq \ \frac{(n-1)(n+2)}{n^2} \ \ (\to 1),$$
$$p_2 \leq p_0^2 \iff \mathbf{E}[L_n] \ \leq \ \frac{(n-1)(5n+6)}{4n^2} \ \ \left(\to \tfrac{5}{4}\right).$$

Both estimates themselves hold for every $n \geq 2$. The first is strict for $n \geq 3$, while the second is strict for $n \geq 4$.

Corollary D3. $\lim_{n\to\infty} \mathbf{E}_{\mathrm{MST}(K_n)}[\deg(x)^2] = 10 - 4\zeta(3) = 5.1917723873616\ldots$, which answers [19, Question 3.8] affirmatively and identifies the limit with the second moment $\mathbf{E}[N^2]$ of the root degree of the wired minimal spanning forest on the Poisson-weighted infinite tree computed in [19, Proposition 3.11].

Proposition D1 is an exact finite identity, and Corollary D2 turns the whole complete-graph problem into a two-sided estimate of $\mathbf{E}[L_n]$; the two halves turn out to be of very different quality (§6). The identity also explains why the same constant $10 - 4\zeta(3)$ appears in the PWIT calculation of [19]: the finite complete-graph second moment is tied exactly to the classical minimum-spanning-tree weight. Corollary D3 and its rate use the asymptotic results of [6, 13]; Theorems B1 and B2 remain finite and effective at every n.

1.4 Identically distributed weights

Question 3.13 is posed for a single common weight law. Since $\mathbf{P}_{\mathrm{MST}}$ itself does not depend on which atomless law is used, one might expect that hypothesis to be cosmetic. It is not: as soon as different edges may carry different laws, no universal constant exists, already on K_4.

Theorem C. For each integer $t \geq 1$ let G_t be the complete graph on $\{a, b, c, d\}$ in which the two disjoint edges ab and cd carry the law with distribution function x^t on $[0, 1]$ (that is, the law of $\max(U_1, \dots, U_t)$) and the four remaining edges are uniform on $[0, 1]$, all six weights independent. Write $R(t)$ for the ratio $p_{ab,cd}/(p_{ab}\, p_{cd})$ of the resulting minimum spanning tree measure. Then

$$R(t) = \frac{2(5t+6)(t+1)(t+4)^2(2t+3)}{9\,(5t^2+12t+8)^2}, \qquad R(t) = \frac{4t}{45} + \frac{46}{75} + O(t^{-1}),$$

and $R(t)$ is strictly increasing on the positive integers, first exceeding 8 at $t = 84$. In particular $R(t) \to \infty$, so no inequality of the form of Question 3.13 holds, with any constant, for independent non-identically distributed atomless edge weights.

The two models are different, and Remark 4.5 quantifies the difference: the natural way to realise a $\max(U_1, \dots, U_t)$ law inside an i.i.d. model, namely subdividing an edge into a path, does not preserve the event "one fixed edge lies in the tree", and the resulting attenuation destroys exactly the amplification that Theorem C produces.

1.5 The idea of the proofs

Theorem A. The obstruction (1) rules out a universal factor 1. Freeze the weights of all edges other than e and f — call this the *environment* — and let a, b, c be the conditional probabilities of $\{e \in T\}$, $\{f \in T\}$ and $\{e, f \in T\}$ given it, so that these are functions of the environment alone. By Kruskal's rule the edge e is accepted exactly when its two endpoints are still separated by the strictly lighter edges. At frozen environment this makes the indicator of $\{e \in T\}$ nonincreasing in $w(e)$ and nondecreasing in $w(f)$, and symmetrically for f: raising the weight of one marked edge helps the other. Two functions of two independent coordinates that are monotone in opposite directions have nonpositive covariance, so Harris's inequality gives $c \leq ab$ pointwise, with no further hypothesis on the tie-free environment. Conditional negative correlation therefore holds almost surely, and the whole failure exhibited by (1) lives in the covariance of a and b under the environment measure.

The remaining term is the environmental covariance. Its natural proxy is the bottleneck distance A between the endpoints of e in the environment graph $H = G - \{e, f\}$: were f absent, e would be accepted precisely when $w(e) < A$, so a would equal A exactly. The presence of f perturbs this by at most a factor 2 in each direction (Lemma 3.4), so with B the bottleneck distance between the endpoints of f it suffices to bound $\mathbf{E}[AB] \leq \sqrt{\mathbf{E}[A^2]\mathbf{E}[B^2]}$, Cauchy–Schwarz absorbing whatever dependence A and B have. What remains is to compare $\mathbf{E}[A^2]$ with $(\mathbf{E}A)^2$, and here the probabilistic content is that A is a percolation first-passage time: $\mathbf{P}(A > t)$ is the probability that two vertices are not connected in Bernoulli(t) percolation on H, and realising density $s + t$ as two independent sprinkled layers shows that this tail is submultiplicative. A nonnegative random variable with submultiplicative tail — a new-better-than-used random variable — satisfies $\mathbf{E}[A^2] \leq 2(\mathbf{E}A)^2$. Three factors of 2, one from that moment bound and one from each of the two perturbation estimates, produce the constant 8.

The complete graph. Here the mechanism is different and the answer is exact. Give every edge of K_n an independent rate-one exponential clock, run Kruskal in the order in which the clocks ring, and record only the accepted mergers. The resulting process on partitions of the vertex set is the discrete skeleton of the multiplicative coalescent: from components of sizes $a_1, \dots, a_k$ the next accepted edge joins two of them with probability proportional to $a_i a_j$, and its two endpoints are uniform in their components, independently of the past. Uniformity of the endpoints is what makes the degree functional computable: if $\Phi = \sum_x \deg(x)(\deg(x) - 1)$ counts ordered pairs of adjacent tree edges, then a merger of blocks of sizes a, b raises $\mathbf{E}\Phi$ by $8 - 4/a - 4/b$, so that $\mathbf{E}[\Phi \mid \text{merger history}] = 8(n-1) - 4H$ with H the sum of $1/a + 1/b$ over the $n - 1$ mergers. Both pair probabilities on K_n are therefore governed by the single functional H, adjacent negative correlation asking for it to be large and disjoint negative correlation for it to be small. A pathwise identity — every component other than the last one is the child of exactly one merger — rewrites H as $n - 1/n + J$, where J sums $1/|C|$ over the components C created by mergers; since a created component has at most n elements, $J \geq (n-1)/n$ pathwise, and that one line is Theorem B1. The other direction needs J bounded above, no pathwise bound is available (Remark 5.7), and this is where the multiplicative merger law enters: a one-step convexity inequality bounds the multiplicatively weighted average of $1/(a_i + a_j)$ over pairs of blocks by $k/(2n)$, and summing over the $n - 1$ states visited gives $\mathbf{E}[J] \leq (n-1)(n+2)/(4n)$, which is Theorem B2. Finally, in continuous time H accumulates at rate $n(\kappa - 1)$, where κ is the number of components, while $L_n = \int_0^\infty (\kappa(t) - 1)\, dt$; so $\mathbf{E}[H] = n\,\mathbf{E}[L_n]$, which is Proposition D1.

2 Preliminaries

2.1 The cycle characterisation

We use one description of the minimum spanning tree throughout. For a real u let $G_{<u}$ denote the subgraph of G consisting of the edges of weight strictly less than u.

Lemma 2.1 (Kruskal / cycle property)**.** *Suppose all edge weights of G are distinct. Then for every edge g with endpoints x, y,*

$$g \in \mathrm{MST}(G) \iff x \text{ and } y \text{ lie in different components of } G_{<w(g)}.$$

In particular a loop is never in $\mathrm{MST}(G)$, and a bridge always is.

Proof. This is the standard correctness statement for Kruskal's algorithm, which processes edges in increasing weight and accepts exactly those joining two distinct current components; with distinct weights the algorithm's output is the unique minimum spanning tree, and the set of edges present when g is processed is exactly $E(G_{<w(g)})$. For a loop, $x = y$ always lie in the same component. For a bridge g, $G_{<w(g)} \subseteq G - g$ never connects x to y. □

Since $\mathrm{MST}(G)$ depends on the weight vector only through its order, and since applying the common distribution function F_μ to every weight is an almost surely strictly order-preserving map onto independent Uniform$[0, 1]$ weights, we may and do assume in §3 that all weights are uniform on $[0, 1]$. In §5 we instead use Exp(1) weights, again without changing $\mathbf{P}_{\mathrm{MST}}$.

2.2 Connection times

Definition 2.2. Let K be a finite multigraph with weights in $[0,1]$ and let x, y be vertices. Set

$$T_K(x,y) = \inf\{t \in [0,1] : \; x \leftrightarrow y \text{ in } K_{<t}\},$$

with the convention $T_K(x,y) = 1$ if x and y lie in different components of K, and $T_K(x,x) = 0$.

Equivalently $T_K(x,y)$ is the bottleneck distance from x to y, the minimum over paths of the largest weight on the path (and 1 if there is no path). Definition 2.2 is the threshold relevant to adding a fresh edge: by Lemma 2.1, if a new edge xy of weight u is added to K then, outside the null set of ties, it is accepted precisely when $u < T_K(x,y)$.

3 A universal factor for all finite multigraphs

Throughout this section the weights are i.i.d. Uniform$[0,1]$. We begin with the estimate that ultimately pays for the correlation, and which is the only place where a percolation argument is used.

3.1 A second-moment bound for connection times

Lemma 3.1. *Let K be a finite multigraph with i.i.d. uniform weights, let x, y be vertices, put $T = T_K(x,y)$ and $q(t) = \mathbf{P}(T > t)$, extended by $q(t) = 0$ for $t \geq 1$. Then*

$$q(s+t) \leq q(s)q(t) \quad (s,t \geq 0), \qquad \textit{and consequently} \qquad \mathbf{E}[T^2] \leq 2(\mathbf{E}T)^2. \tag{3}$$

Proof. For $s+t \geq 1$ the left side vanishes, so assume $s+t<1$. For $u \in [0,1]$ the subgraph $K_{<u}$ is Bernoulli(u) percolation on $E(K)$, and $q(u)$ is the probability that x and y are not connected in it. Realise Bernoulli$(s+t)$ percolation as the union of two independent layers of densities s and $r = t/(1-s)$: an edge is absent from the union with probability $(1-s)(1-r) = 1-s-t$, as required. If the union fails to connect x to y, then each layer separately fails to connect them. The two layers are independent, q is nonincreasing, and $r \geq t$; hence

$$q(s+t) \leq q(s)q(r) \leq q(s)q(t).$$

This also covers the case in which x, y lie in different components of K, where $q \equiv 1$ on $[0,1)$. For the second assertion, all integrals below are over $[0,\infty)$ and q vanishes on $[1,\infty)$, so by Tonelli

$$(\mathbf{E}T)^2 = \iint q(s)q(t)\,ds\,dt \;\geq\; \iint q(s+t)\,ds\,dt = \int u\,q(u)\,du = \tfrac{1}{2}\,\mathbf{E}[T^2]. \qquad \square$$

The submultiplicativity in (3) is the two-layer sprinkling decomposition behind the classical square-root trick, and its consequence is the standard second-moment inequality for a new-better-than-used random variable; see e.g. [3].

Remark 3.2. The constant 2 in (3) cannot be improved. If K consists of k parallel xy-edges then T is the minimum of k uniforms, so $\mathbf{E}T = 1/(k+1)$, $\mathbf{E}[T^2] = 2/((k+1)(k+2))$ and

$$\frac{\mathbf{E}[T^2]}{(\mathbf{E}T)^2} = \frac{2(k+1)}{k+2} \;\nearrow\; 2.$$

3.2 Freezing the environment

Fix distinct edges $e \neq f$ of G, write $H = G - \{e, f\}$ for the multigraph on the same vertex set with e and f deleted, and put $U = w(e)$, $V = w(f)$. Let $\mathcal{F} = \sigma\big(w(g) : g \in E(H)\big)$ be the σ-field generated by the *environment*, i.e. by all weights other than those of the two marked edges. For a tie-free realisation W of the environment, which occurs almost surely, set

$$a(W) = \mathbf{P}(e \in T \mid \mathcal{F} = W), \quad b(W) = \mathbf{P}(f \in T \mid \mathcal{F} = W), \quad c(W) = \mathbf{P}(e, f \in T \mid \mathcal{F} = W), \tag{4}$$

where the conditional probabilities average only over the independent uniform pair (U, V).

Lemma 3.3 (conditional negative correlation)**.** *For every tie-free environment W,*

$$c(W) \leq a(W)\, b(W).$$

Proof. Fix W and regard the acceptance indicators $\mathbf{1}_{\{e \in T\}}$ and $\mathbf{1}_{\{f \in T\}}$ as functions of $(U, V) \in [0, 1]^2$. By Lemma 2.1, e is accepted precisely when its endpoints are not connected by the edges of weight $< U$, that is by

$$\{g \in E(H) : w(g) < U\} \ \cup \ \{f \text{ if } V < U\}.$$

Increasing U only enlarges this edge set, so $\mathbf{1}_{\{e \in T\}}$ is nonincreasing in U; increasing V can only remove f from it, so $\mathbf{1}_{\{e \in T\}}$ is nondecreasing in V. By symmetry $\mathbf{1}_{\{f \in T\}}$ is nondecreasing in U and nonincreasing in V.

Now change coordinates to $(U, 1 - V)$, which are again independent and uniform. In these coordinates $\mathbf{1}_{\{e \in T\}}$ is coordinatewise nonincreasing and $\mathbf{1}_{\{f \in T\}}$ is coordinatewise nondecreasing. Harris's inequality [11] on a product of two totally ordered probability spaces gives nonpositive covariance for a decreasing and an increasing function, which is exactly $c(W) \leq a(W)b(W)$. $\square$

The conditioning is essential: by (1), the unconditional failure lives in the covariance of the two functions a and b. We compare them with connection times in the graph with both marked edges deleted.

3.3 A deletion sandwich

Lemma 3.4. *Let e have endpoints x_1, x_2 and f have endpoints y_1, y_2, and set*

$$A = A(W) = T_H(x_1, x_2), \qquad B = B(W) = T_H(y_1, y_2)$$

for the connection times in the deleted environment H (Definition 2.2). Then, for every tie-free W,

$$\tfrac{A}{2} \ \leq \ A - \tfrac{A^2}{2} \ \leq \ a(W) \ \leq \ A, \qquad \tfrac{B}{2} \ \leq \ B - \tfrac{B^2}{2} \ \leq \ b(W) \ \leq \ B.$$

Proof. Upper bound. Adding f to H can only connect x_1 to x_2 at a smaller or equal threshold, so if e is accepted then, outside the null set $\{U = A\}$, we have $U < A$; the set $\{U < A\} \subseteq [0, 1]^2$ has area A.

Lower bound. If $U < \min(A, V)$ then f is not among the edges of weight $< U$, and H alone does not connect x_1 to x_2 below U because $U < A$; so e is accepted by Lemma 2.1. The area of $\{U < \min(A, V)\}$ is

$$\int_0^1 \min(A, v)\, dv = \frac{A^2}{2} + A(1 - A) = A - \frac{A^2}{2} \ \geq \ \frac{A}{2},$$

the last step because $A \leq 1$. The statements for f follow by exchanging the roles of e and f. $\square$

3.4 Proof of Theorem A

We combine conditional negative correlation, the deletion sandwich, and the connection-time moment bound.

Proof of Theorem A. By §2 we may take the weights uniform on $[0, 1]$. With the notation of (4) and Lemma 3.4, and taking all expectations with respect to the environment,

$$\mathbf{P}_{\mathrm{MST}}(e, f \in T) = \mathbf{E}[c] \overset{\text{L.3.3}}{\leq} \mathbf{E}[ab] \overset{\text{L.3.4}}{\leq} \mathbf{E}[AB] \leq \sqrt{\mathbf{E}[A^2]\,\mathbf{E}[B^2]} \overset{\text{L.3.1}}{\leq} 2\,\mathbf{E}[A]\,\mathbf{E}[B],$$

where the third step is Cauchy–Schwarz and the fourth applies Lemma 3.1 separately to the two terminal pairs $\{x_1, x_2\}$ and $\{y_1, y_2\}$ in the multigraph H with its i.i.d. uniform weights. Finally Lemma 3.4 gives $\mathbf{E}[A] \leq 2\mathbf{E}[a] = 2\,\mathbf{P}_{\mathrm{MST}}(e \in T)$ and likewise $\mathbf{E}[B] \leq 2\,\mathbf{P}_{\mathrm{MST}}(f \in T)$, so

$$\mathbf{P}_{\mathrm{MST}}(e, f \in T) \leq 2 \cdot 2\,\mathbf{P}_{\mathrm{MST}}(e \in T) \cdot 2\,\mathbf{P}_{\mathrm{MST}}(f \in T) = 8\,\mathbf{P}_{\mathrm{MST}}(e \in T)\,\mathbf{P}_{\mathrm{MST}}(f \in T). \qquad \square$$

The same estimates improve the constant when both marked edges are rare.

Corollary 3.5 (small marginals). *Put $g(p) = (1 - \sqrt{1 - 4p})/2$. If $p_e, p_f \leq 1/4$, then*

$$p_{ef} \leq 2g(p_e)g(p_f).$$

If in addition $p_e p_f > 0$, then

$$\frac{p_{ef}}{p_e p_f} \leq 2\frac{g(p_e)}{p_e}\frac{g(p_f)}{p_f}.$$

Consequently the upper bound on the ratio is $2 + o(1)$ as $\max(p_e, p_f) \to 0$.

Proof. Write $m_A = \mathbf{E}[A]$ and $m_B = \mathbf{E}[B]$. Lemmas 3.4 and 3.1 give

$$p_e = \mathbf{E}[a] \geq m_A - \tfrac{1}{2}\mathbf{E}[A^2] \geq m_A - m_A^2, \qquad p_e \geq \tfrac{1}{2}m_A,$$

and likewise for f. If $p_e \leq 1/4$, the second inequality gives $m_A \leq 1/2$, and inversion of the increasing function $m \mapsto m - m^2$ on $[0, 1/2]$ gives $m_A \leq g(p_e)$. The proof of Theorem A already gives $p_{ef} \leq 2m_A m_B$, which proves the assertions; finally $g(p)/p \to 1$ as $p \downarrow 0$. $\square$

3.5 Degenerate configurations

No simplicity or 2-connectivity hypothesis has entered, and the degenerate cases are covered rather than excluded. If e is a loop then $x_1 = x_2$, so $A = 0$, $a \equiv 0$ and $p_e = p_{ef} = 0$, and the inequality reads $0 \leq 0$. If e is a bridge of G then H does not connect x_1 to x_2, so $A = 1$ and $p_e = 1$; the chain above is valid, and one should expect no strictness, since two bridges give $p_{ef} = p_e p_f = 1$. If e and f have the same endpoints then $p_{ef} = 0$, because the heavier of the two closes a cycle with the lighter, and Lemma 3.3 remains valid. Finally $H = G - \{e, f\}$ need not be connected; Definition 2.2 and Lemma 3.1 were set up to cover this ($T = 1$), and Lemma 3.4 is pointwise, so nothing changes.

4 Explicit correlation examples

We first quantify the parallel-bundle construction of [15], then give a simple-graph counterexample to pairwise negative correlation, and finally show why the common-law hypothesis of Theorem A is necessary. The two K_4 families use the following four-terminal calculation.

4.1 A four-terminal formula

Label the vertices of K_4 by a, b, c, d and distinguish the two disjoint edges

$$\mathsf{A} = ab, \qquad \mathsf{B} = cd.$$

The remaining four edges ac, ad, bc, bd form a complete bipartite graph between $\{a, b\}$ and $\{c, d\}$; give them a common continuous distribution function H, and give A, B the continuous distribution functions F, G, all six weights independent. Write $\bar{H} = 1 - H$.

Lemma 4.1. *With the above notation,*

$$p_{\mathsf{A}} = \int \Big[(1 - G(x))(1 - H(x)^2)^2 + G(x)\{2\bar{H}(x)^2 - \bar{H}(x)^4\}\Big]\, dF(x), \tag{5}$$

$$p_{\mathsf{B}} = \textit{the same with } (F, \mathsf{A}) \textit{ and } (G, \mathsf{B}) \textit{ interchanged}, \tag{6}$$

$$\begin{aligned} p_{\mathsf{AB}} = &\iint_{x<y} \bar{H}(y)^2 \big[2(1 - H(x)^2) - \bar{H}(y)^2\big]\, dF(x)\, dG(y) \\ &+ \iint_{y<x} \bar{H}(x)^2 \big[2(1 - H(y)^2) - \bar{H}(x)^2\big]\, dF(x)\, dG(y). \end{aligned} \tag{7}$$

Proof. For (5), condition on $w(\mathsf{A}) = x$; by Lemma 2.1, A is accepted iff $a \not\leftrightarrow b$ below x.

If $w(\mathsf{B}) > x$, which has probability $1 - G(x)$, only cross edges are present below x, and $a \leftrightarrow b$ below x iff c receives both of ac, bc below x, or d receives both of ad, bd below x. These two events are independent, each of probability $H(x)^2$, so the probability of non-connection is $(1 - H(x)^2)^2$.

If $w(\mathsf{B}) < x$, of probability $G(x)$, then c and d are already merged, and $a \leftrightarrow b$ below x iff a has at least one cross edge below x *and* b has at least one; those events are independent with probability $1 - \bar{H}(x)^2$ each. The probability of non-connection is $1 - (1 - \bar{H}(x)^2)^2 = 2\bar{H}(x)^2 - \bar{H}(x)^4$.

For (7), take the ordering $x = w(\mathsf{A}) < y = w(\mathsf{B})$ and write $q = 1 - H(x)^2$, $v^2 = \bar{H}(y)^2$. For each $u \in \{c, d\}$ let N_u be the event that u has no cross edge below y (probability v^2) and B_u the event that u does not receive both of its cross edges below x (probability q); $N_u \subseteq B_u$ since $x < y$, and the pairs of events at c and at d are independent. As above, A is accepted iff $B_c \cap B_d$ occurs. Given that A has been accepted and lies below y, the vertices a, b are merged, and B is accepted iff at most one of c, d is attached to that merged block below y, i.e. iff $N_c \cup N_d$ occurs. Hence

$$\mathbf{P}(\mathsf{A}, \mathsf{B} \text{ accepted} \mid x, y) = \mathbf{P}\big((B_c \cap B_d) \cap (N_c \cup N_d)\big) = v^4 + 2v^2(q - v^2) = 2qv^2 - v^4,$$

which is the first integrand of (7). The other order is symmetric. □

4.2 Parallel bundles, and a lower bound for the optimal constant

A natural attempt to enlarge the ratio is to iterate the construction of [15], enlarging the two parallel bundles. It does not work, and the failure is quantitative. Let $R_{r,s}$ denote the ratio for one marked copy in each of two disjoint parallel bundles of sizes r, s on K_4, the four cross edges uniform. Regard each bundle as a macro-edge whose weight is its minimum. If a macro-edge is selected, its unique lightest copy is selected, and symmetry therefore divides its marginal by r or s and the joint probability by rs; these factors cancel in the ratio. Substituting the laws of

$\min(U_1, \ldots, U_r)$ and $\min(V_1, \ldots, V_s)$ into Lemma 4.1, and then applying these symmetry factors, gives, writing $D = (r+s+2)(r+s+3)(r+s+4)$ and $I(r,s) = \frac{4}{(s+2)(r+s+3)} - \frac{3s+10}{(s+2)(s+4)(r+s+4)}$,

$$p_e = \frac{r+6}{(r+2)(r+4)} + \frac{4}{D}, \qquad p_f = \frac{s+6}{(s+2)(s+4)} + \frac{4}{D}, \qquad p_{ef} = I(r,s) + I(s,r).$$

Proposition 4.2. *For all positive integers r, s,*

$$R_{r,s} = \frac{p_{ef}}{p_e p_f} \ \le\ \frac{78100}{77841} = 1.003327\ldots,$$

with equality exactly at $r = s = 4$. In the symmetric subfamily,

$$R_{t,t} = \frac{(t+1)^2(t+4)(2t+3)(2t^2+19t+34)}{(2t^3+17t^2+34t+22)^2},$$
$$R_{t,t} - 1 = \frac{20t^3+9t^2-78t-76}{(2t^3+17t^2+34t+22)^2} = \frac{5}{t^3} + O(t^{-4}),$$

so the ratio peaks at $t = 4$ and returns to 1. The configuration of [15] is $(r,s) = (3,3)$, with $R_{3,3} = 109872/109561$.

Proof. Put $m = r+s$ and $\rho = rs$, and define

$$D = (m+2)(m+3)(m+4), \qquad K = (m+2)^2(m+3)(m+4),$$
$$L = D + 24, \qquad C = 6D + 32,$$
$$N_0 = K(6m^3 + 78m^2 + 356m + 544), \qquad N_1 = K(m^2 + 7m + 16),$$
$$D_0 = 4Cm^2 + LCm + C^2, \qquad D_1 = L^2 + 4Lm - 8C.$$

Simplifying the three displayed probabilities gives

$$R_{r,s} = \frac{N_0 + N_1\rho}{D_0 + D_1\rho + 16\rho^2}. \tag{8}$$

For fixed m, the derivative with respect to ρ has the sign of

$$Q_m(\rho) = N_1 D_0 - N_0 D_1 - 32N_0\rho - 16N_1\rho^2,$$

and $Q'_m(\rho) = -32N_0 - 32N_1\rho < 0$ for $\rho \ge 0$. The feasible range is $m - 1 \le \rho \le \lfloor m^2/4 \rfloor$.

For $2 \le m \le 10$, direct substitution at the right endpoint gives

$$\begin{aligned}(Q_m(\lfloor m^2/4 \rfloor))_{m=2}^{10} = (&623232000, 3726475200, 15482880000, 50485284864,\\ &133754572800, 297101597760, 541023436800,\\ &751512453120, 464792380416),\end{aligned}$$

so R is increasing throughout the feasible interval. Its balanced endpoint values, in the same order, are

$$\frac{44}{45}, \frac{3745}{3774}, \frac{840}{841}, \frac{14385}{14362}, \frac{109872}{109561}, \frac{763587}{761105}, \frac{78100}{77841}, \frac{385385}{384154}, \frac{83772}{83521}.$$

The largest is the $m = 8$ value.

For $m \geq 11$, put $u = m - 11$. At the left endpoint,

$$\begin{aligned}-Q_m(m-1) = {} & 930527357760 + 3158322926304u + 1989328507888u^2 \\ & + 609067397832u^3 + 112783678672u^4 + 13814498848u^5 \\ & + 1164466368u^6 + 68281008u^7 + 2747600u^8 + 72576u^9 \\ & + 1136u^{10} + 8u^{11} > 0.\end{aligned}$$

Thus R is decreasing on the feasible interval. If Num and Den denote the numerator and denominator of (8) at $\rho = m - 1$, then

$$\begin{aligned}78100\,\mathrm{Den} - 77841\,\mathrm{Num} = {} & 34642293840 + 28712778516u + 9667374224u^2 \\ & + 1607247273u^3 + 144551244u^4 + 7147790u^5 \\ & + 181300u^6 + 1813u^7 > 0.\end{aligned}$$

This proves the bound for $m \geq 11$ as well. Equality in the full argument requires $m = 8$ and $\rho = 16$, hence $r = s = 4$. Substitution of $r = s = t$ in (8) gives the two symmetric formulas, and $t = 3$ gives the values from [15]. □

Thus the parallel-bundle route cannot push the ratio above 1.0034. The simple graphs below exceed this family maximum. Appendix A describes an independent exact check of the formulas and certificate.

4.3 Positive correlation in a simple graph

Parallel edges are not necessary for the failure of pairwise negative correlation.

Proposition 4.3. *Let G be the simple graph on $\{a, b, c, d, z\}$ with*

$$E(G) = \{ab, ac, ad, az, bc, bd, bz, cd\},$$

and mark $e = az$ and $f = cd$. Under i.i.d. atomless edge weights,

$$p_e = \frac{7}{12}, \qquad p_f = \frac{69}{140}, \qquad p_{ef} = \frac{145}{504}, \qquad \frac{p_{ef}}{p_e p_f} = \frac{1450}{1449} > 1.$$

Proof. All 8! strict edge orders are equally likely. Running Kruskal's rule through those orders accepts e in 23520 orders, accepts f in 19872 orders, and accepts both in 11600 orders. Dividing by $8! = 40320$ gives the three stated probabilities and hence the ratio. Appendix A records a second exact enumeration by relative-order patterns. □

This witness is minimal by edge count. Indeed, a smallest counterexample may be assumed bridgeless: a bridge belongs to every spanning tree, and deleting all bridges separates independent minimum-spanning-tree problems. A connected bridgeless simple graph with m edges has at most m vertices. An exhaustive exact census of all such graphs with $m \leq 7$ finds no violation; the census is recorded in Appendix A.

The smallest witness is not the strongest one we know. Let S be a set of hub vertices with no internal edges, take four further vertices a, b, c, d, and join every vertex of S to all four of a, b, c, d. Add the two marked edges $A = ab$ and $B = cd$.

Proposition 4.4. *For the preceding simple graph, two hubs give*

$$p_A = p_B = \frac{1}{2}, \qquad p_{AB} = \frac{1186}{4725}, \qquad \frac{p_{AB}}{p_A p_B} = \frac{4744}{4725}.$$

Three hubs give

$$p_A = p_B = \frac{1123}{2730}, \qquad p_{AB} = \frac{71479}{420420}, \qquad \frac{p_{AB}}{p_A p_B} = \frac{13938405}{13872419} = 1.004756\ldots.$$

In particular the second ratio is larger than the maximum 78100/77841 *in Proposition 4.2.*

Proof. For two hubs, full enumeration of the 10! strict edge orders gives marginal counts 1814400 and joint count 910848. For three hubs, exact summation over the relative positions of the two marked edges and the other twelve edges gives marginal count 35861253120 and joint count 14821885440 out of 14! orders. Dividing gives the stated probabilities. The relative-position calculation is also applied to the two-hub graph and agrees with its full enumeration. □

4.4 Non-identical laws: proof of Theorem C

The second specialisation replaces the parallel bundles, which are a device for producing a min of uniforms, by their opposite.

Proof of Theorem C. Take $F(x) = G(x) = x^t$ and $H(x) = x$ on $[0,1]$ in Lemma 4.1. All integrands are then polynomials and the integrals are elementary; carrying them out gives

$$p_{\mathsf{A}} = p_{\mathsf{B}} = A_t := \frac{3(5t^2+12t+8)}{(t+1)(t+2)(t+4)(2t+3)}, \qquad p_{\mathsf{AB}} = J_t := \frac{2(5t+6)}{(t+1)(t+2)^2(2t+3)}. \tag{9}$$

Therefore

$$R(t) = \frac{J_t}{A_t^2} = \frac{2(5t+6)(t+1)(t+4)^2(2t+3)}{9\,(5t^2+12t+8)^2},$$

a rational function whose numerator has degree 5 and whose denominator has degree 4, with leading coefficients 20 and 225. Hence $R(t) \to \infty$ and $R(t)/t \to 20/225 = 4/45$; expanding one order further gives the constant 46/75.

For monotonicity write $R(t) = 2N(t)/(9D(t))$, omitting the displayed constant factors from the numerator and denominator. The sign of $R(t+1) - R(t)$ is the sign of $Q(t) = N(t+1)D(t) - N(t)D(t+1)$. With $u = t-1$, exact expansion gives

$$\begin{aligned} Q(u+1) = {}& 124000 + 669300u + 1289264u^2 + 1262269u^3 + 712831u^4 \\ &+ 242311u^5 + 48935u^6 + 5400u^7 + 250u^8, \end{aligned}$$

which is positive for $u \geq 0$. Direct substitution gives $R(83) < 8 < R(84)$, so 84 is the first crossing. □

The exact values $R(1) = 44/45$, $R(2) = 168/169$, $R(3) = 8232/7921$, $R(4) = 2860/2601$ are immediate from the displayed formula, and $R(3000) = 267.28\ldots$ Note that $t = 1$ is plain K_4 with all six weights uniform, so $R(1) = 44/45$ is the disjoint-pair ratio p_2/p_0^2 at $n = 4$ and agrees with Table 1.

Remark 4.5 (why this does not contradict Theorem A). The law $\max(U_1, \ldots, U_t)$ arises naturally inside an i.i.d. model: if an edge is subdivided into a path of t edges with private internal vertices and i.i.d. uniform weights, then, after subtracting the constant total path weight, the minimum spanning tree of the quotient is Kruskal on the base graph with the path *maxima* as effective weights. So the family of Theorem C is realisable at the level of *macro-edges*. It is not realisable at the level of a single fixed edge, and the loss is exactly quantifiable. Let X, Y be the indicators that the two macro-edges lie in the quotient tree, with probabilities $\mathbf{P}(X) = p$ and $\mathbf{P}(Y) = q$, and let I_1, I_2 be one fixed constituent edge in each path, both paths of length t. A constituent edge is omitted exactly when its macro-edge is omitted *and* it carries the maximum of its own path; the identity of that maximum is uniform along the path and independent of its value, so

$$\mathbf{P}(I_1) = 1 - \frac{1-p}{t} =: \alpha, \qquad \mathbf{P}(I_2) = 1 - \frac{1-q}{t} =: \beta, \qquad \mathrm{Cov}(I_1, I_2) = \frac{\mathrm{Cov}(X,Y)}{t^2},$$

for all p, q. If $p, q > 0$ and the macro-edge relative excess is nonzero, this identity may be divided by that excess to give

$$\frac{\mathbf{P}(I_1, I_2)/(\mathbf{P}(I_1)\mathbf{P}(I_2)) - 1}{\mathbf{P}(X,Y)/(pq) - 1} = \frac{pq}{t^2\alpha\beta} \;\le\; \frac{1}{t^2},$$

the last step because $\alpha \ge p$ and $\beta \ge q$. Applied to (9) this leaves the fixed-edge ratio $1 + 5t^{-5} + O(t^{-6})$, so the amplification is destroyed. Theorem C is a theorem about the independent non-identical model, and Theorem A is a theorem about the i.i.d. model; the distinction is essential.

5 The complete graph

5.1 The accepted-merger coalescent

Give every edge of K_n an independent rate-one exponential clock and run Kruskal's algorithm in the order in which the clocks ring. This does not change $\mathbf{P}_{\mathrm{MST}}$, since the tree depends only on the weight order, and it has the advantage that the weights are now the ring times, so the total weight of the tree is L_n.

Discard the rejected clocks and record only the *accepted mergers*. There are exactly $n-1$ of them, and they merge the vertex set from n singletons to one block.

Lemma 5.1. *At time zero, or just after any accepted merger other than the final one, condition on the whole past and let $A_1, \ldots, A_k$ be the current components, of sizes $a_1, \ldots, a_k$; thus $k \ge 2$. Then the next accepted merger joins A_i and A_j with probability*

$$\frac{a_i a_j}{Z}, \qquad Z = \sum_{i<j} a_i a_j = \frac{n^2 - \sum_i a_i^2}{2}, \tag{10}$$

and, conditionally on that pair, the two endpoints of the accepted edge are independent and uniform in A_i and A_j respectively, and independent of the past.

Proof. An edge of K_n whose endpoints lie in two distinct current components has not yet rung: had it rung earlier, Kruskal would have accepted it at that time and merged those components. By memorylessness of the exponential the residual clocks of all such edges are i.i.d. Exp(1) given the past, so the next accepted edge is uniform among the Z cross edges. Uniformity among the $a_i a_j$ edges between A_i and A_j is exactly the assertion that, given the pair, the endpoints are independent and uniform; and this is independent of the past, which involves only clocks that have already rung. □

Lemma 5.1 identifies the sequence of component sizes with the discrete skeleton of the multiplicative coalescent, cf. [2]. This is the only place where the complete graph enters. Appendix A checks the transition model against a direct evaluation of $\mathbf{P}_{\mathrm{MST}}$.

The two functionals below are the bookkeeping devices for the whole section.

Definition 5.2. A *merger history* is the sequence of pairs of component sizes merged, in order. For a merger history on n leaves put

$$H = \sum_{\text{mergers}} \Big(\frac{1}{a} + \frac{1}{b}\Big), \qquad J = \sum_{\text{mergers}} \frac{1}{a+b},$$

where a, b are the sizes of the two components merged (the *children*) and $a+b$ is the size of the component created (the *parent*).

Lemma 5.3 (pathwise accounting)**.** *For every merger history on n leaves,*

$$H = n - \frac{1}{n} + J.$$

Proof. Every component other than the final one of size n is a child of exactly one merger, so $H = \sum_{C \neq \text{root}} 1/|C|$, the sum being over all components ever present. The components ever present are the n singletons together with the $n-1$ components created by mergers, and J sums $1/|C|$ over the created ones. Hence

$$H = \Big(n + J\Big) - \frac{1}{n},$$

the term n from the singletons and the correction $-1/n$ from removing the root, which is created but is nobody's child. □

5.2 The degree functional

Let the accepted forest carry the degree function deg and set

$$\Phi = \sum_{x \in V} \deg(x)\big(\deg(x) - 1\big),$$

so that in the final tree Φ counts the ordered pairs of distinct tree edges sharing a vertex.

Lemma 5.4. *Conditionally on the entire merger history,*

$$\mathbf{E}[\Phi \mid \textit{merger history}] = 8(n-1) - 4H. \tag{11}$$

Proof. If a merger joins components of sizes a, b through endpoints u, v, then Φ increases by $\big[(\deg(u)+1)\deg(u) - \deg(u)(\deg(u)-1)\big] + \big[\text{same at } v\big] = 2\deg(u) + 2\deg(v)$. Each component is a tree, so the degrees inside a component of size a sum to $2(a-1)$; by Lemma 5.1 the endpoint u is uniform in its component and independent of everything else, whence $\mathbf{E}[\deg(u) \mid \cdot] = 2(a-1)/a$. Therefore

$$\mathbf{E}[\Delta\Phi \mid \cdot] = \frac{4(a-1)}{a} + \frac{4(b-1)}{b} = 8 - \frac{4}{a} - \frac{4}{b}.$$

The endpoint choices are independent of the sequence of component sizes, since the evolution (10) depends on sizes only; so we may condition on the whole merger history and sum over the $n-1$ mergers, obtaining $8(n-1) - 4H$ by Definition 5.2. □

Identity (11) reduces both pair probabilities on K_n to the single functional H, and by Lemma 5.3 to J. Adjacent negative correlation needs J bounded below, disjoint negative correlation needs it bounded above, and the two directions require quite different handles on the same quantity.

5.3 Adjacent pairs: proof of Theorem B1

Proof of Theorem B1. Every parent size in a merger history is at most n, and there are $n-1$ mergers, so pathwise

$$J = \sum_{\text{mergers}} \frac{1}{a+b} \;\geq\; \frac{n-1}{n}, \tag{12}$$

with strict inequality for $n \geq 3$ because the first merger has parent size $2 < n$. By Lemma 5.3, $H \geq n - 1/n + (n-1)/n = n + (n-2)/n$ pathwise, and hence by Lemma 5.4

$$\mathbf{E}[\Phi] \;\leq\; 8(n-1) - 4n - \frac{4(n-2)}{n} = \frac{4(n-1)(n-2)}{n}, \tag{13}$$

strictly for $n \geq 3$. By vertex exchangeability, if D is the degree of a fixed vertex then $\mathbf{E}[D(D-1)] = \mathbf{E}[\Phi]/n$. Finally, K_n has exactly $n(n-1)(n-2)$ ordered pairs of distinct edges sharing a vertex (choose the shared vertex, then an ordered pair of distinct other endpoints), and each contributes p_1 to $\mathbf{E}[\Phi]$, so

$$p_1 = \frac{\mathbf{E}[\Phi]}{n(n-1)(n-2)} < \frac{1}{n(n-1)(n-2)} \cdot \frac{4(n-1)(n-2)}{n} = \frac{4}{n^2} = p_0^2. \qquad \square$$

The adjacent case thus rests on a single pathwise observation: a component created by a merger has at most n elements.

5.4 Disjoint pairs: proof of Theorem B2

For the other direction we need J bounded *above* in expectation, and here the trivial estimate is useless: parent sizes can be as small as 2. The upper bound is where the multiplicative merger law (10) is used, and it comes from a convexity inequality applied one step at a time.

Lemma 5.5 (one-step partition inequality)**.** *Let $k \geq 2$ and let $a_1, \ldots, a_k$ be positive integers with $\sum_i a_i = n$. Then*

$$\sum_{i<j} \frac{a_i a_j}{a_i + a_j} \;\leq\; \frac{k}{2n} \sum_{i<j} a_i a_j. \tag{14}$$

Equality holds if and only if $k = 2$, or $a_1 = \cdots = a_k$.

Proof. Write $S_2 = \sum_i a_i^2$ and $Z = \sum_{i<j} a_i a_j = (n^2 - S_2)/2$. From the identity

$$\frac{ab}{a+b} = \frac{a+b}{4} - \frac{(a-b)^2}{4(a+b)}$$

and from $a_i + a_j \le n$ we get

$$\sum_{i<j} \frac{a_i a_j}{a_i + a_j} \;\le\; \frac{1}{4}\sum_{i<j}(a_i + a_j) - \frac{1}{4n}\sum_{i<j}(a_i - a_j)^2.$$

Now $\sum_{i<j}(a_i + a_j) = (k-1)n$ and $\sum_{i<j}(a_i - a_j)^2 = (k-1)S_2 - 2Z = kS_2 - n^2$, so the right-hand side equals

$$\frac{(k-1)n}{4} - \frac{kS_2 - n^2}{4n} = \frac{k(n^2 - S_2)}{4n} = \frac{kZ}{2n},$$

which is (14).

For the equality case, the only inequality used was $(a_i - a_j)^2/(a_i + a_j) \ge (a_i - a_j)^2/n$, which is an equality for the pair $\{i, j\}$ exactly when $a_i = a_j$ or $a_i + a_j = n$. If $k = 2$ then $a_1 + a_2 = n$ and equality holds. If $k \ge 3$ and two parts differ, say $a_i \ne a_j$, then equality forces $a_i + a_j = n$, which is impossible since the remaining $k - 2 \ge 1$ parts are positive. Hence for $k \ge 3$ equality holds exactly when all parts are equal, and in that case both sides equal $\binom{k}{2} n/(2k)$. □

Since the left side of (14) is, up to the normalisation Z, exactly the conditional expectation of the increment of J, summing over the states visited turns it into the bound we need.

Lemma 5.6 (reciprocal budget)**.** *For the accepted-merger coalescent on K_n,*

$$\mathbf{E}[J] \;\le\; \frac{(n-1)(n+2)}{4n}, \qquad \textit{hence} \qquad \mathbf{E}[H] \;\le\; \frac{(n-1)(5n+6)}{4n}, \tag{15}$$

with strict inequality in both for $n \ge 4$, and equality for $n = 2, 3$.

Proof. By Lemma 5.1, from a state with component sizes $a_1, \dots, a_k$ the increment of J has conditional expectation

$$\mathbf{E}[\Delta J \mid \text{state}] = \frac{1}{Z}\sum_{i<j} \frac{a_i a_j}{a_i + a_j} \;\le\; \frac{k}{2n}$$

by Lemma 5.5. The process visits a state with exactly k components once for each $k = n, n-1, \dots, 2$, and J is the sum of the increments at those states, so by the tower property

$$\mathbf{E}[J] \;\le\; \sum_{k=2}^{n} \frac{k}{2n} = \frac{n(n+1)/2 - 1}{2n} = \frac{(n-1)(n+2)}{4n}.$$

The second bound in (15) follows from Lemma 5.3:

$$\mathbf{E}[H] = n - \frac{1}{n} + \mathbf{E}[J] \le \frac{4n^2 - 4 + (n-1)(n+2)}{4n} = \frac{(n-1)(5n+6)}{4n}.$$

For strictness, note that for $n \ge 4$ the state reached after the first merger is $(2, 1, 1, \dots, 1)$ with $k = n - 1 \ge 3$ components which are not all equal, so Lemma 5.5 is strict there. For $n = 2$ and $n = 3$ every state visited has $k = 2$ or is the all-ones state, so every step is an equality; indeed $J \equiv 5/6$ and $H \equiv 7/2$ deterministically at $n = 3$. □

Remark 5.7 (the merger law is needed, not only the state space). Unlike (12), Lemma 5.6 has no pathwise version. For the balanced merger history on $n = 4$ leaves — merge two disjoint pairs of singletons, then merge the two blocks — we have

$$J = \frac{1}{2} + \frac{1}{2} + \frac{1}{4} = \frac{5}{4} \;>\; \frac{(n-1)(n+2)}{4n} = \frac{9}{8},$$

so by (11) that history has $\mathbf{E}[\Phi \mid \text{history}] = 4$, below the threshold $3(n-1)(n-2)/n = 4.5$ that the proof below requires. Nor would the state space alone suffice: replacing the multiplicative choice (10) by a uniform choice among the $\binom{k}{2}$ pairs of blocks breaks Lemma 5.5 at once. At the state $(1, 1, 2)$, where $n = 4$, $k = 3$ and the bound $k/(2n)$ is $3/8$, the multiplicatively weighted average $Z^{-1} \sum_{i<j} a_i a_j/(a_i + a_j)$ equals $11/30 \le 3/8$, while the unweighted average of $1/(a_i + a_j)$ is $7/18 > 3/8$.

Proof of Theorem B2. Let $n \ge 4$. Combining (11) with Lemma 5.6,

$$\mathbf{E}[\Phi] = 8(n-1) - 4\,\mathbf{E}[H] \;>\; 8(n-1) - \frac{(n-1)(5n+6)}{n} = \frac{3(n-1)(n-2)}{n}. \tag{16}$$

The tree has $(n-1)(n-2)$ ordered pairs of distinct edges, of which Φ are adjacent; hence it has $(n-1)(n-2) - \Phi$ ordered disjoint pairs. The number of ordered pairs of disjoint edges of K_n is $6\binom{n}{4} = n(n-1)(n-2)(n-3)/4$, and each contributes p_2, so

$$p_2 = \frac{4\big[(n-1)(n-2) - \mathbf{E}[\Phi]\big]}{n(n-1)(n-2)(n-3)} \;<\; \frac{4(n-1)(n-2)\big[1 - 3/n\big]}{n(n-1)(n-2)(n-3)} = \frac{4(n-3)}{n^2(n-3)} = \frac{4}{n^2} = p_0^2. \qquad \square$$

Proof of Corollary B3. Two distinct edges of K_n either share a vertex or not; apply Theorem B1 or Theorem B2. For $n = 3$ no disjoint pair exists. $\square$

Remark 5.8. In the language of [19, Corollary 2.4], which the counting in this section and Corollary D2 reprove: with D the degree of a fixed vertex, $p_1 \le p_0^2 \iff \mathbf{E}[D^2] \le 6 - 14/n + 8/n^2$ and $p_2 \le p_0^2 \iff \mathbf{E}[D^2] \ge 5 - 11/n + 6/n^2$. So the two halves of Corollary B3 squeeze $\mathbf{E}[D^2]$ into a window of width $1 - 3/n + 2/n^2$ around 5–6; the limiting window is $(5, 6)$ and the true limit is $10 - 4\zeta(3) = 5.1917\ldots$ by Corollary D3.

6 The identity, and the reduction to $\mathbf{E}[L_n]$

The bounds of §5 were two estimates of the same functional H. That functional has an exact meaning, and identifying it is what turns the pair correlations on K_n into a statement about a classical quantity. Recall that L_n is the weight of $\mathrm{MST}(K_n)$ under i.i.d. $\mathrm{Exp}(1)$ weights, so that in the coalescent of §5 the weights are the clock ring times.

Lemma 6.1. $\mathbf{E}[H] = n\,\mathbf{E}[L_n]$ *for every* $n \ge 2$.

Proof. Let $\kappa(t)$ be the number of components of the graph formed by the edges of weight at most t. Since the weight of an edge is its ring time,

$$L_n = \sum_{g \in T} w(g) = \int_0^\infty \#\{g \in T : w(g) > t\}\, dt = \int_0^\infty (\kappa(t) - 1)\, dt,$$

because at time t the number of accepted mergers still to come is exactly $\kappa(t)-1$. In the continuous-time process, at a state with components of sizes $a_1,\dots,a_k$ the total accepted-merger rate is Z and the expected rate at which the functional H accumulates is

$$\sum_{i<j} a_i a_j\Big(\frac{1}{a_i}+\frac{1}{a_j}\Big)=\sum_{i<j}(a_i+a_j)=n(k-1),$$

while $\kappa-1=k-1$. Thus $H_t-n\int_0^t(\kappa(s)-1)\,ds$ is a martingale. Since $H\le 2(n-1)$ and L_n is integrable, letting $t\to\infty$ gives $\mathbf{E}[H]=n\,\mathbf{E}[L_n]$. □

Proof of Proposition D1. With D the degree of a fixed vertex, $\mathbf{E}[D(D-1)]=\mathbf{E}[\Phi]/n$ by exchangeability and $\mathbf{E}[D]=2(n-1)/n$ because every spanning tree has $n-1$ edges. Hence, by (11) and Lemma 6.1,

$$\mathbf{E}[D^2]=\frac{\mathbf{E}[\Phi]}{n}+\frac{2(n-1)}{n}=\frac{8(n-1)-4n\mathbf{E}[L_n]}{n}+\frac{2(n-1)}{n}=\frac{10(n-1)}{n}-4\,\mathbf{E}[L_n]. \qquad \square$$

Proof of Corollary D2. For $n\ge 3$, Proposition D1 and (11) show that $p_1\le p_0^2$ is $\mathbf{E}[\Phi]\le 4(n-1)(n-2)/n$, i.e. $8(n-1)-4n\mathbf{E}[L_n]\le 4(n-1)(n-2)/n$, i.e. $\mathbf{E}[L_n]\ge(n-1)(n+2)/n^2$. For $n\ge 4$, likewise $p_2\le p_0^2$ is $\mathbf{E}[\Phi]\ge 3(n-1)(n-2)/n$, i.e. $\mathbf{E}[L_n]\le(n-1)(5n+6)/(4n^2)$. The lower estimate follows for every $n\ge 2$ from (12), and the upper estimate from Lemma 5.6; their equality statements follow from the strictness statements there. □

Thus Theorem B1 is the statement $\mathbf{E}[L_n]\ge(n-1)(n+2)/n^2$ and Theorem B2 is the statement $\mathbf{E}[L_n]\le(n-1)(5n+6)/(4n^2)$, and §5 proves both without any input from the minimum-spanning-tree-weight literature. Numerically the lower bound tends to 1 and is far from tight, while the upper bound tends to 5/4 and is tight to about 1%: see Table 1. Thus the adjacent case, which requires the lower bound on $\mathbf{E}[L_n]$, is resolved here by the looser of the two estimates. Both pair ratios increase in the computed range. The maximum of $\mathbf{E}[L_n]$ in Gamarnik's range $2\le n\le 45$ occurs at $n=8$, where $\mathbf{E}[L_8]=1.2459812\ldots$ against the upper bound $161/128=1.2578125$; decrease for every $n\ge 8$ remains conjectural [10, Conjecture 4.1].

Proposition D1 also recasts Gamarnik's conjectured decrease: for $n\ge 8$, $\mathbf{E}[L_n]>\mathbf{E}[L_{n+1}]$ is equivalent to

$$\mathbf{E}[\deg_{n+1}(x)^2]-\mathbf{E}[\deg_n(x)^2]>\frac{10}{n(n+1)},$$

where $\deg_k$ is the degree in $\mathrm{MST}(K_k)$.

Remark 6.2 (what Theorem B2 says about the expected tree weight). Corollary D2 exposes Theorem B2 as the non-asymptotic bound

$$\mathbf{E}[L_n]\ \le\ \frac{(n-1)(5n+6)}{4n^2}\qquad(n\ge 2), \tag{17}$$

a statement about a quantity with a long history: Frieze [9], Steele [18], Janson [12], Fill and Steele [8], Gamarnik [10], Nishikawa, Otto and Starr [17], Cooper, Frieze, Ince, Janson and Spencer [6], and Li and Zhang [13]. These works develop asymptotics, exact small-n evaluations, and polynomial or Tutte representations. None of these sources states the all-n upper bound (17). It is exact at $n=2,3$ and lies within 0.0119 of the largest value in Gamarnik's computed range, but its elementary form makes prior knowledge plausible; Theorem B2 may therefore be a new complete-range proof of a known bound rather than a new inequality. Theorem B1 uses the other half of Corollary D2 and is unaffected.

n	p_1/p_0^2	p_2/p_0^2	$\mathbf{E}[L_n]$	lower bd. $\frac{(n-1)(n+2)}{n^2}$	upper bd. $\frac{(n-1)(5n+6)}{4n^2}$
3	0.7500000000	—	1.1667	10/9	7/6
4	0.7555555556	0.9777777778	1.2167	9/8	39/32
5	0.7597552910	0.9804894180	1.2353	28/25	31/25
6	0.7630719281	0.9825707626	1.2427	10/9	5/4
7	0.7657742674	0.9842257326	1.2454	54/49	123/98
8	0.7680285888	0.9855771290	1.2460	35/32	161/128
9	0.7699442296	0.9867038469	1.2455	88/81	34/27
10	0.7715965139	0.9876591349	1.2445	27/25	63/50
11	0.7730392810	0.9884803595	1.2432	130/121	305/242
12	0.7743121839	0.9891945849	1.2418	77/72	121/96
13	0.7754451621	0.9898219351	1.2405	180/169	213/169
14	0.7764613009	0.9903777088	1.2391	52/49	247/196
20	0.7809063153	0.9927279258	1.2323	209/200	1007/800
30	0.7850808025	0.9948028441	1.2250	232/225	377/300

Table 1: Exact pair correlations for $\mathrm{MST}(K_n)$ and the two-sided estimate of $\mathbf{E}[L_n]$ from Corollary D2. Every entry comes from an exact rational; the $\mathbf{E}[L_n]$ column is rounded to four decimals. The full exact table for $3 \le n \le 30$ accompanies the paper.

Proof of Corollary D3. The expansion of Cooper, Frieze, Ince, Janson and Spencer [6] in particular gives $\mathbf{E}[L_n^{\mathrm{unif}}] \to \zeta(3)$ for i.i.d. Uniform$[0,1]$ weights. Li and Zhang [13] give the quantitative transfer

$$\mathbf{E}[L_n] - \mathbf{E}[L_n^{\mathrm{unif}}] = \frac{\zeta(3)}{n} + O\Big(\frac{\log^2 n}{n^2}\Big), \tag{18}$$

so $\mathbf{E}[L_n] \to \zeta(3)$ for exponential weights as well. Insert this into Proposition D1:

$$\lim_{n\to\infty} \mathbf{E}_{\mathrm{MST}(K_n)}[\deg(x)^2] = 10 - 4\zeta(3) = 5.1917723873616\ldots$$

By [19, Proposition 3.11] this is exactly $\mathbf{E}[N^2]$ for the root degree N of the wired minimal spanning forest on the Poisson-weighted infinite tree [16], which is the assertion of [19, Question 3.8]. $\square$

The sharper expansion identifies the rates and the two limiting correlation ratios. Let c_1, c_2 be the constants in [6, Theorem 1], so that

$$\mathbf{E}[L_n^{\mathrm{unif}}] = \zeta(3) + \frac{c_1}{n} + \frac{c_2 + o(1)}{n^{4/3}}, \qquad c_1 = 0.0384956\ldots. \tag{19}$$

Corollary 6.3 (asymptotic rates)**.** *For exponential weights,*

$$\mathbf{E}[L_n] = \zeta(3) + \frac{c_1 + \zeta(3)}{n} + \frac{c_2 + o(1)}{n^{4/3}},$$
$$\mathbf{E}[\deg(x)^2] = 10 - 4\zeta(3) - \frac{10 + 4(c_1 + \zeta(3))}{n} - \frac{4c_2 + o(1)}{n^{4/3}}.$$

Moreover,

$$\frac{p_1}{p_0^2} = 2 - \zeta(3) + \frac{4 - c_1 - 4\zeta(3)}{n} - \frac{c_2 + o(1)}{n^{4/3}},$$
$$\frac{p_2}{p_0^2} = 1 + \frac{4\zeta(3) - 5}{n} + O(n^{-2}).$$

In particular the degree second moment converges at order n^{-1}, while the adjacent and disjoint ratios tend to $2 - \zeta(3)$ and 1, respectively.

Proof. The first line follows by combining (18) and (19); Proposition D1 gives the second. If $m_n = \mathbf{E}[\deg(x)^2]$, direct counting of ordered adjacent and disjoint edge pairs gives

$$\frac{p_1}{p_0^2} = \frac{m_n - 2 + 2/n}{4(1 - 3/n + 2/n^2)} \quad (n \geq 3), \qquad \frac{p_2}{p_0^2} = \frac{n^2((n-1) - m_n)}{(n-1)(n-2)(n-3)} \quad (n \geq 4).$$

Substitution and expansion prove the last two formulas. □

This argument does not pass through the local weak limit. The local weak limit of $\mathrm{MST}(K_n)$ is due to Addario-Berry [1], and convergence in distribution of the root degree does not by itself give convergence of the second moment, which is why Question 3.8 was asked; the recent study of dynamic local convergence for Prim's algorithm [7] likewise does not address moments. Proposition D1 replaces the missing uniform integrability by an exact finite identity. Corollary 6.3 gives an asymptotic rate, although the $o(1)$ in (19) does not supply explicit finite error constants. Tang and Zhang [19, Proposition 3.11] print the decimal 5.191717; the correct value of $10 - 4\zeta(3)$ is $5.1917723873616\ldots$, and the closed form there is not affected.

7 Open problems

Theorem A locates positive correlation in the covariance of two bottleneck distances, while Proposition D1 controls the complete-graph pair correlations through the expected tree weight. Both descriptions leave natural quantitative questions.

Problem 7.1 (the optimal constant)**.** Determine the least C for which the conclusion of Theorem A holds. We know only that it lies in $[13938405/13872419, 8]$. Corollary 3.5 gives the sharper factor $2 + o(1)$ when both marginals vanish, but does not establish a global factor 2. The factor 8 comes from the sharp inequality $\mathbf{E}[T^2] \leq 2(\mathbf{E}T)^2$ and one factor 2 from each deletion sandwich; improving it requires using more of the joint structure than these three separate bounds.

Problem 7.2 (the simple-graph constant)**.** Determine the supremum of $p_{ef}/(p_e p_f)$ over finite connected simple graphs. Proposition 4.4 gives the lower bound 13938405/13872419, while Theorem A gives the upper bound 8.

Problem 7.3 (effective errors and monotonicity)**.** Corollary 6.3 gives the asymptotic rate and the limits of both complete-graph ratios, but not explicit finite error constants. Can the error in (19) be made effective enough to give useful finite-n bounds? Both ratios increase for $3 \leq n \leq 30$ where defined (Table 1); are they monotone throughout their domains?

Problem 7.4 (other families at every size)**.** Tang and Zhang [19, Remark 1.2] state that their method also gives the disjoint-pair property for $K_{n,n}$ for all sufficiently large n. Is there an all-n analogue of Corollary B3 for $K_{n,n}$, or for other natural families? The method of §5 uses the complete-graph transition rate (10) and does not transfer as it stands. The minimum spanning tree weight has been studied on several other host graphs, for instance regular graphs [4], so the analogue of Corollary D2 there is a concrete starting point. Finally, is there an analogue of Theorem A for the wired minimal spanning forest on an infinite graph?

A Exact computations and computer assistance

The universal statements of §3 and the complete-graph theorems of §5–6 use no computation. Computation enters in three places: it supplies the exact census of Table 1, proves the finite enumerations in Propositions 4.3 and 4.4, and checks the algebra of Proposition 4.2 and the transition model of Lemma 5.1 against direct evaluations of $\mathbf{P}_{\mathrm{MST}}$. Every quantity is an exact rational; decimals are rounded only when printed. The full table for $3 \le n \le 30$ accompanies the paper as `data/kn-exact-table.csv`. The code uses only the Python standard library, and the repository records the commands and provenance.

A.1 What each computation evaluates

1. *Brute force over orderings.* For a small multigraph, enumerate all $m!$ orderings of the edges, run Kruskal on each, and count. This evaluates $\mathbf{P}_{\mathrm{MST}}$ directly from its definition. It is used on K_4 (720 orders), the five-vertex graph of Proposition 4.3 (40320 orders), the two-hub graph of Proposition 4.4 (10! orders), and small multigraphs.

2. *Exact summation over order patterns.* Condition on the set of edges lighter than e, or, for p_{ef}, on the three blocks into which the two marked weights split the remaining $m-2$ edges, and sum the exact rational probability of each pattern. This too evaluates $\mathbf{P}_{\mathrm{MST}}$ directly, by a different enumeration, and reaches K_5, both simple-graph propositions, and the example (1).

3. *The accepted-merger coalescent.* The dynamic program over integer partitions given by (10), returning $\mathbf{E}[J]$, $\mathbf{E}[H]$, $\mathbf{E}[\Phi]$, $\mathbf{E}[D^2]$, p_1, p_2 and $\mathbf{E}[L_n]$ exactly. This is the machinery of §5, so it is not independent evidence for §5; it is what routes 1 and 2 are used to test.

4. $\mathbf{E}[L_n]$ *from the area identity.* Independently of the coalescent, $\mathbf{E}[L_n] = \int_0^\infty (\mathbf{E}[\kappa(t)] - 1)\, dt$ where $\kappa(t)$ counts components of $G(n, 1-e^{-t})$; substituting $u = e^{-t}$ turns the integrand into an integer polynomial in u built from the exact connectivity polynomial of $G(s,p)$, so the integral is a finite rational sum. This route knows nothing about mergers, degrees, or the coalescent.

Lemma 5.1 is checked from outside by these routes: routes 1 and 2 agree with each other on every marginal and every pair probability of K_4, and both agree with route 3 on p_1, p_2, on $\mathbf{E}[\Phi] = 8(n-1) - 4\mathbf{E}[H]$ and on Proposition D1 at $n = 4$ and $n = 5$; route 4 agrees with route 3 on $\mathbf{E}[L_n]$; and route 3 agrees with the published table of [10] through $n = 30$ in the full suite (the committed transcription extends through $n = 45$). The self-consistency checks $p_0 = 2/n$, $\sum_e p_e = n-1$ and $\mathbf{E}[H] = n - 1/n + \mathbf{E}[J]$ are also verified, the last recomputed from the dynamic program rather than assumed.

A.2 What is verified, and in what range

The lemmas of §5 are verified exhaustively in finite ranges: Lemma 5.3 and the pathwise bound (12) on every merger history shape up to $n = 12$; Lemma 5.5, together with its equality characterisation, on every integer partition of every $n \le 32$; and the closing algebra of §5 and §6 as rational identities for $n < 300$. The chain of inequalities in the proof of Theorem A is verified, instance by instance and exactly, on an explicitly generated exhaustive universe of connected labelled multigraphs — including loops, parallel classes and disconnected deleted environments — for every marked pair, together with the frozen-environment forms of Lemmas 3.3 and 3.4 obtained by exact integration over the two marked weights. The bundle family of Proposition 4.2 is evaluated for all $r+s \leq 120$ in the widest documented run, and its two positive-coefficient certificates are re-expanded coefficient by coefficient. The positive-coefficient certificate for $R(t+1) > R(t)$ is likewise re-expanded; an additional finite scan reaches $t = 300$ in the widest run. The two- and three-hub probabilities are recomputed by relative-order enumeration, and the two-hub graph also by all 10! orders. The minimality statement after Proposition 4.3 uses the bridge reduction printed there and a census of 1528 labelled bridgeless graphs with at most seven edges, collapsing to 17 isomorphism classes and 272 representative marked pairs. Finally, the formulas of Theorem C and Remark 4.5 are checked directly on the corresponding i.i.d. subdivided graphs for $t = 1, 2, 3$.

Several tempting strengthenings fail, and the suite retains them as controls: $C = 1$ fails by (1); $\mathbf{E}[T^2] \leq \frac{3}{2}(\mathbf{E}T)^2$ fails for the minimum of six uniforms; $a \geq \frac{3}{4}A$ fails when the marked edges join the same two vertices and no other edge is present; Lemma 3.3 is an equality when both marked edges are bridges; and Lemma 5.5 fails at $(2, 1, 1)$ if block pairs are chosen uniformly rather than with multiplicative weights.

These finite checks do not replace the proofs above. They test the modelling and algebra and establish the explicitly finite assertions: the exact table, the enumerated counterexamples, and the edge-count minimality census.

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